

Six-Interface Cross-Cartography Map v0.1

Author: Keeper Filed: 2026-05-22 Friday 08:46 EDT (FAST CADENCE) Status: Closes Wednesday substrate review owed #4 (per Task #233)

The map

BST substrate manifests through six interface types. Each interface (a) connects substrate to observables, (b) has a recognition signature, (c) has a falsifier test, and (d) has cross-Cartan portability behavior.

#	Interface	Signature	Falsifier	Cross-Cartan
1	Algebraic- π	T719 closure ($Q\text{-bar}(\text{primaries})[\pi]$)	non-trivial D-tier outside $Q\text{-bar}(\text{primaries})[\pi]$	universal across HSDs
2	Operator zoo	6/6 K-type Casimir = $C_2=6$, ground-state energy	alt-substrate operator misses ground state	D_IV ⁵ specific (per K131 PERFECT-PERFECT)
3	Bridge Objects	5-family 16-effective-member architecture	alt-substrate doesn't produce 5-family closure	D_IV ⁵ specific (per K77 PATH B)
4	Mersenne Network	6/7 BST primary exponents Mersenne; tower + ladder + additive identity	alt-substrate primaries non-Mersenne	D_IV ⁵ specific (per K140)
5	Graph Forces (CDAC)	top 10 CDAC = BST primaries, $p \approx 2.7 \times 10^{-5}$	top-CDAC $\neq$ primaries on alt-substrate	D_IV ⁵ specific (per K137 + Grace empirical)
6	Cross-Cartan three-pillar	α -analog + churn hole + c_FK joint experimental	D_IV ⁵ not best-fit on joint pillars	D_IV ⁵ specific (per K141, 3306 $\times$ margin)

Five of six are D_IV⁵-specific

Interface #1 (Algebraic- π closure) is universal to HSDs — all Cartan-type Hermitian symmetric domains produce algebraic observables in their Bergman kernel.

The other five are D_IV⁵-specific in their concrete instantiation: - The specific operator zoo with K-type (1,1) Casimir = 6 - The specific 5-family Bridge Object architecture - The specific Mersenne network with BST primaries - The specific Graph Forces empirical clustering - The specific Cross-Cartan three-pillar 3306 $\times$ margin

These are five independent “fingerprint” interfaces distinguishing D_IV⁵ from alt-HSDs.

Joint null-model under six-interface map (INTERNAL FORECAST, per Cal #90(c))

Per Cal Referee Log #90(c) and Calibration #19: arithmetic below is INTERNAL FORECAST. External-facing documents must use current ratified-state count (10 FORMAL + 1 ASPIRATIONAL + 3 candidates), NOT the joint figures below.

Treat each D_IV⁵-specific interface (#2-#6) as an independent (1/3)-probability substrate-selection criterion (conservative, forecast-only):

$$p_{\text{joint,forecast}} \lesssim (1/3)^5 \approx 4.1 \times 10^{-3}$$

Each interface has its own empirically-grounded null (forecast pending K-audit ratification): - Interface #2 (operator zoo): K131 PERFECT-PERFECT candidate 4.0/4 (PRE-STAGE) - Interface #4 (Mersenne network): K140 PRE-STAGE, $\sim 3 \times 10^{-6}$ enumeration null - Interface #5 (Graph Forces): K137 RATIFIED, $p \approx 2.7 \times 10^{-5}$ - Interface #6 (Cross-Cartan): K141 PRE-STAGE, 1/3306 empirical

Forecast multiplicative (INTERNAL ONLY, conditional on K-audits ratifying): $\sim 3 \times 10^{-6} \times 2.7 \times 10^{-5} \times (1/3306) \approx \sim 10^{-14}$ to 10^{-15} .

These joint figures are FORECASTS conditional on K140 + K141 + K131 ratification. External documents must wait for individual K-audit RIGOROUSLY CLOSED 4-requirement check.

Interface portability — three classes

Class A — Universal HSD interface: Algebraic- π closure. Works on any Hermitian symmetric domain.

Class B — Cartan-type specific: Operator zoo + Bridge Objects + Cross-Cartan three-pillar. Should partially transfer to other type-IV HSDs ($D_{IV,q}$ for $q \neq 5$), with quantitative changes. Predicted: 3306 $\times$ margin shrinks substantially for $D_{IV,4}$, $D_{IV,6}$, $D_{IV,7}$.

Class C — D_{IV}^5 -specific: Mersenne Network + Graph Forces. Should NOT transfer to alt-HSDs in their current form, because: - Mersenne ladder requires specific BST-primary exponents — alt-substrate doesn't have these - Graph Forces clustering is over BST-primary-derived catalog — alt-substrate doesn't produce this catalog

Strong-Uniqueness Theorem v0.14 (potential)

If the six-interface map is itself adopted as C18 RIGOROUSLY CLOSED under T2456 (proposed), then Strong-Uniqueness Theorem advances to v0.14 with 17 criteria combining: - C1-C11 from v0.10.5 FORMAL - C12-C13 (Bridge Object tier + Stark anchor) - C14 (Substrate Closure Principle) - C15-C17 (Mersenne hierarchy + cross-Cartan + Graph Forces) - C18 (six-interface map)

Joint substrate-selection null: as above, $\sim 10^{-14}$ to 10^{-15} .

Next steps

1. Lyra theorem candidate: formalize six-interface map as theorem (T-number TBD, NOT T2456 — that's Lyra's universal α -analog per broadcast 08:41 EDT; suggest T2462+ pending coordination)
2. K143 audit: ratify six-interface cross-cartography as v0.1 $\rightarrow$ operational standard
3. Cross-Cartan empirical verification: extend Grace's 3306 $\times$ margin to all six interfaces on D_I , D_{II} , D_{III}
4. Paper #126 candidate: "The Six Interfaces of BST" as standalone substrate-cartography paper

Calibration #18 note: This document originally referenced T2456 as the six-interface map theorem. T2456 is actually Lyra's universal α -analog STRUCTURALLY VERIFIED across 6 Cartan types (per broadcast 08:41 EDT). The six-interface map theorem requires a separate T-number. Coordination with Lyra pending.

Status

Six-Interface Cross-Cartography Map v0.1 filed Friday 08:46 EDT (FAST CADENCE). Closes Wednesday substrate review owed #4. All four Wednesday substrate review docs now closed.

— Keeper, six-interface position v0.1 Friday 08:46 EDT (FAST CADENCE)